

Shailza Jolly

✉ shailzajolly@gmail.com

☎ 0176 67279750

📍 Berlin, Germany

🌐 shailzajolly

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Summary

- Senior ML Scientist/Engineer with a Ph.D. in CS and 6+ years spanning academic research and industry. Specializes in LLMs, Generative AI, and scalable data pipelines. Brings hands-on depth in modern LLM architectures, efficiency techniques (KV-cache, speculative decoding, FlashAttention), and agentic system design, alongside a track record of top-tier publications (AAAI, NAACL, EMNLP) and production delivery.

Education

Ph.D.	TU Kaiserslautern , Computer Science	Kaiserslautern, Germany
	- Grade: Sehr Gut 1.0 (highest on 1.0–5.0 scale) - Thesis: Building Natural Language Generation and Understanding Systems in Data-Constrained Settings - Work on VQA, interpretability, and conversational AI	2019 – 2022
Visiting Researcher	University of Copenhagen , Natural Language Processing	Copenhagen, Denmark
	Developed an unsupervised post-editing algorithm to generate fluent fact-check-explanations	Jan 2021 – Mar 2021
M.Sc.	TU Kaiserslautern , Computer Science — Minor in Economics	Kaiserslautern, Germany
	Grade: Sehr Gut 1.5	2016 – 2018
Semester Abroad	Kyushu University , Computer Science	Fukuoka, Japan
	Explainable AI to analyze behavior of deep CNN architectures for image recognition	Nov 2017 – Feb 2018
B.Tech	Guru Nanak Dev Engineering College , Computer Science & Engineering	India
	Grade: First Division with Distinction	2012 – 2016

Experience

Career Break / Independent Study	Generative AI & LLM Systems	Berlin, Germany
	- Deepened expertise in modern LLM architectures and efficiency — FlashAttention, RoPE, scaling laws, KV-cache, speculative decoding, and fast inference techniques. - Studied agentic system design (tool use, MCP) and evaluation strategies for LLM/agentic systems; built small prototypes to test concepts.	July 2025 – present 9 months
Flinn.ai	Senior AI Engineer	Berlin, Germany
	- Led development of AI-driven product features including data extraction from medical research papers and multilingual complaint monitoring. - Partnered with product and backend teams to scope problems, define success metrics, and deliver end-to-end features within an agile product lifecycle. - Role eliminated due to company-wide strategic shift.	Jan 2025 – June 2025 6 months
Parental Leave (Maternity)		Berlin, Germany
	Stayed engaged with research literature and maintained technical skills.	Dec 2023 – Dec 2024 1 year 1 month

Amazon AI , Research Scientist - Led development of a scalable noise-removal/data-quality pipeline processing billions of tokens to improve training data for LLMs. - Wrote research plans and technical documentation; presented results and recommendations to senior science/engineering leadership. - Mentored Master's/Ph.D. interns and collaborated on research deliverables.	Berlin, Germany July 2022 – Nov 2023 1 year 5 months
German Research Center for Artificial Intelligence (DFKI) , Machine Learning Scientist - Delivered ML research and engineering in BMBF-funded projects XAINES (Explainable AI) and DeFuseNN (vision-language systems). - Supervised interns and BSc/MSc students; provided technical mentorship and project guidance.	Berlin, Germany Mar 2019 – June 2022 3 years 4 months
NVIDIA Research , Machine Learning Scientist Intern Built a document understanding pipeline combining table/cell detection, tabular structure retrieval, and OCR for financial documents.	Santa Clara, US May 2021 – Aug 2021 4 months
Amazon Alexa , Applied Scientist Intern Developed a method for generating diverse synthetic training data to improve intent classification and slot labeling for task-oriented NLU.	Aachen, Germany Aug 2019 – Dec 2019 5 months
SAP AI Research , Research Intern / Master's Thesis Designed and implemented an evaluation metric for Visual Question Answering (VQA) models.	Berlin, Germany May 2018 – Feb 2019 10 months

Selected Publications

Search and Learn: Improving Semantic Coverage for Data-to-Text Generation Shailza Jolly, Z. X. Zhang, A. Dengel, L. Mou arxiv.org/abs/2112.02770 (AAAI)	Jan 2022
EaSe: A Diagnostic Tool for VQA Based on Answer Diversity Shailza Jolly, S. Pezzelle, M. Nabi www.aclweb.org/anthology/2021.naacl-main.192 (NAACL-HLT)	Jan 2021
Can Pre-training Help VQA with Lexical Variations? Shailza Jolly, S. Kapoor www.aclweb.org/anthology/2020.findings-emnlp.257 (Findings of EMNLP)	Jan 2020
Data-Efficient Paraphrase Generation to Bootstrap Intent Classification and Slot Labeling for New Features in Task-Oriented Dialog Systems Shailza Jolly, T. Falke, C. Tirkaz, D. Sorokin www.aclweb.org/anthology/2020.coling-industry.2 (COLING (Industry Track))	Jan 2020
How Do Convolutional Neural Networks Learn Design? Shailza Jolly, B. K. Iwana, R. Kuroki, S. Uchida ieeexplore.ieee.org/abstract/document/8545624 (International Conference on Pattern Recognition (ICPR))	Jan 2018

Awards

AAAI-22 Scholarship by Hitachi Awarded by Hitachi to attend AAAI 2022.	Jan 2022
AI Newcomer 2021 Award Recognized by the German Informatics Society and BMBF, Germany.	Jan 2021

EU-Cost STSM Grant

Jan 2020

- Awarded by EU-Cost Action to work on the Multi3generation project at CopeNLU, University of Copenhagen.

Best Student Paper Award

Jan 2018

- Awarded by the International Conference on Pattern Recognition (ICPR) for "How Do Convolutional Neural Networks Learn Design?"

Skills

ML/GenAI: LLMs, NLU/NLG, multimodal learning, synthetic data, model evaluation

LLM Systems: Agentic workflows (MCP, LangGraph/LangSmith), RAG, grounding, experiment design, offline evaluation

Frameworks & Libraries: Python, PyTorch, Hugging Face (Transformers, PEFT, Datasets), PySpark, Weights & Biases

Search & Retrieval: Sentence Transformers, reranking, vector search, Pinecone, Chroma

Infrastructure & Deployment: AWS, Docker, FastAPI, Git

Media Coverage

- [Interview featured in the Rheinfalz newspaper.](#)
- Radio interview for Antenne Kaiserslautern, Germany.

Languages

English: Fluent

German: Beginner

Hindi: Native speaker